Technical appendix: enrollment uncertainty and the proposed closure of Mesa Elementary

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September 2026 — appendix to a letter to the BVSD Board of Education

Summary

The district proposes an irreversible step on the basis of projections with no stated uncertainty. Its own projections for these two schools have moved 10–16% between annual runs, and one more October count could shift the merged school's expected enrollment by two classrooms in either direction. Keeping Mesa open is cheap and reversible; closing it is not.

The numbers that matter (source pages in parentheses; n is the number of school-year cases behind each statistic):

- Bear Creek's 2029–30 projection rose from 272 to 310 (+14%) and Mesa's fell from 224 to 202 (−10%) between the January 2025 and January 2026 runs; Bear Creek's 2028–29 projection went 248, 274, 288 in three consecutive runs (Feb. 2024 report p. 11; Feb. 2025 report p. 9; Feb. 2026 report p. 9).

- The district's own January 2026 projections for the two schools sum to 521 in 2030–31; its merged-school range is 403–462 (deck p. 51), which implies that only 41–71% of Mesa's projected students would land at Bear Creek.

- Across all BVSD elementary schools, the district's two-year-ahead projections have a 90th-percentile absolute error of 13.0% (n = 29); one year ahead, 9.4% (n = 59). Revisions to targets four to five years out have an 80th-percentile magnitude of about 12% (n = 59 per horizon).

- An independent cohort-survival model, back-tested on 26 schools, gives the merged school a 2030–31 median of 430 under a trend kindergarten specification and 501 under a level specification (90% of Mesa students following). The 71-student gap between the two defensible specifications exceeds the 59-student width of the district's entire range. The probability of exceeding the building's 492 seats is 16% under one and 58% under the other; of exceeding three rounds (450), 38% and 85%.

- One more October count is expected to move the model's central estimate for 2030–31 across a range of about 100 students (10th to 90th percentile of 120 simulated outcomes), about two classrooms in either direction.

- The district states recurring savings of \$3.5–4.0M a year for the whole package against one-time costs of \$7.5–10.0M (facility modifications) plus \$5.0M (transition support), with no per-school figures (deck p. 60).

Pre-empting one objection. The independent model's median at 90% retention (430) falls inside the district's range (403–462). That is not the finding. The finding is the probability mass above 450 and above 492, and the fact that an undisclosed modelling choice about kindergarten intake moves the median by more than the width of the district's range.

Conceded at the outset. Mesa is small; it has been on the district's enrollment-advisory list every year the list has existed; and the district's median projection for it is not unreasonable. The argument is about the evidence bar for an irreversible step, not about whether Mesa's enrollment is low.

The evidence standard

An irreversible decision should be held to a higher bar than a reversible one. Closing a school cannot be undone; keeping one open can. Before approving any closure the Board should have in hand:

1. a projection with stated prediction intervals, back-tested against the district's own October counts;
2. the retention, resident-capture and open-enrollment assumptions behind any merged-school projection, stated explicitly;
3. a demonstration that the merged school fits the receiving building with margin under the upside enrollment cases, not only the median, net of the classroom footprint of center-based special-education programs;
4. a school-specific accounting of one-time costs, recurring savings, and revenue at risk from families who leave the district;
5. a transition plan for RISE, school-age care, preschool and transportation.

Sections 4–11 work the enrollment evidence in full and show that (i)–(iii) are not met by the documents presented. Section 12 shows that (iv) is not met. Section 13 lists the open questions under (v).

Background: the decision and the numbers behind it

On August 25, 2026 the district presented its Resilient Schools Proposal; the Board is scheduled to discuss it on September 8 and act on September 22, with changes effective in 2027–28 (deck p. 3). For South Boulder it proposes to "close Mesa and merge the school communities in the Bear Creek building", to "redraw attendance boundaries to combine the Bear Creek, Mesa and dual attendance areas", and to move Mesa's RISE program to Bear Creek (deck p. 49).

The stated rationale (deck p. 50) is that the consolidation "reduces excess capacity and boosts enrollment at the receiving school to 3-rounds"; that Mesa "is projected to continue to decline" and "has the second lowest number of resident students in its attendance area"; and that Bear Creek is at 63% utilization and 2.1 classes per grade against Mesa's 54% and 1.5. The post-change projection (deck p. 51) is in Table 1.

The proposal's figures for the consolidation (Aug. 25, 2026 deck, p. 51; text layer).

	2025–26 (actual)	2027–28 (proj.)	2030–31 (proj.)
Bear Creek: capacity / residents / enrolled	492 / 275 / 312		
Mesa: capacity / residents / enrolled	418 / 228 / 224		
Consolidated at Bear Creek: resident students		473	522
Consolidated at Bear Creek: enrolled students		392–445	403–462
Consolidated utilization		80–90%	82–94%

The deck's footnote on the range reads: "Projected enrollment ranges dependent on number of resident students from the sending school attending the receiving school" (p. 51). No probability is attached to either end and the retention shares behind the two ends are not stated.

District-defined thresholds used throughout: one "round" is about 150 students, three rounds about 450 (Oct. 21, 2025 work session, slide 8); the Long Range Advisory Committee's advisory status is ≤ 2 rounds and ≤ 60% of capacity, and its community-engagement status ≤ 1.5 rounds and ≤ 50% (Feb. 2026 report p. 10). Mesa has appeared on the advisory list in every year it has been published (Feb. 2024 report p. 13; Feb. 2025 report p. 11).

Data and verification

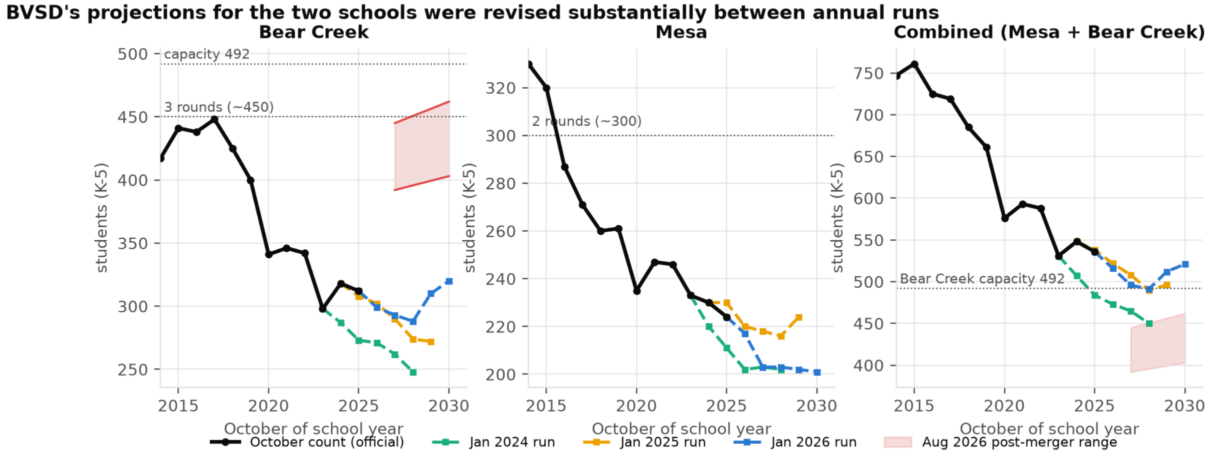
Every number here traces to a page of a BVSD or Colorado Department of Education (CDE) document held unmodified under `data/raw/`. Sources: the Aug. 25, 2026 deck (75 pp.); the February 2024, 2025 and 2026 Annual Enrollment Trend Reports; the Oct. 21, 2025 work session; the March 11 and Sept. 9, 2025 attendance-boundary items; BVSD's October pupil-count files (head count, FTE, special programs) for 2014–15 through 2025–26; CDE pupil-membership files for 2019–20 through 2025–26; the bvsd.org Resilient Schools page and FAQ; and the district's Dec. 12, 2025 enrollment news article.

The capacity tables in the trend reports and work session are images. They were OCR'd; every Mesa and Bear Creek cell was checked by eye against the rendered page, and every other school's row was accepted only if the table's own current-year enrollment matched the official October count (all rows pass; one dropped token was read from the page). Grade counts sum exactly to published head counts for every school in all 12 years. CDE's K–5 membership agrees with BVSD's funded count to within one or two students in every year; CDE additionally records 25 pre-kindergarten students at Mesa in 2025–26 that appear in no BVSD table. Full detail is in `data/VERIFICATION.md`; disagreements between sources in `data/CONFLICTS.md`.

Two figures that circulate publicly differ from the documents. Mesa's October 2025 count is 224 in every BVSD table (CDE's membership count is 225). Bear Creek's "program capacity" is 467 (3.0 rounds) in the January 2024 table and 492 (3.5 rounds) in every later table; the change is not explained in the documents reviewed, and it alone moves Bear Creek's stated utilization by five points.

How much the district's own projections have moved

Figure 1 overlays the three annual runs on the official counts; Table 2 gives the numbers.



Official October counts (black) and BVSD's projections by run; the shaded band is the proposal's post-merger range.

BVSD projections by run and target year (K–5). Feb. 2024 report p. 11; Feb. 2025 report p. 9 (re-printed as Oct. 2025 work session slide 17); Feb. 2026 report p. 9; official counts from the October pupil-count files.

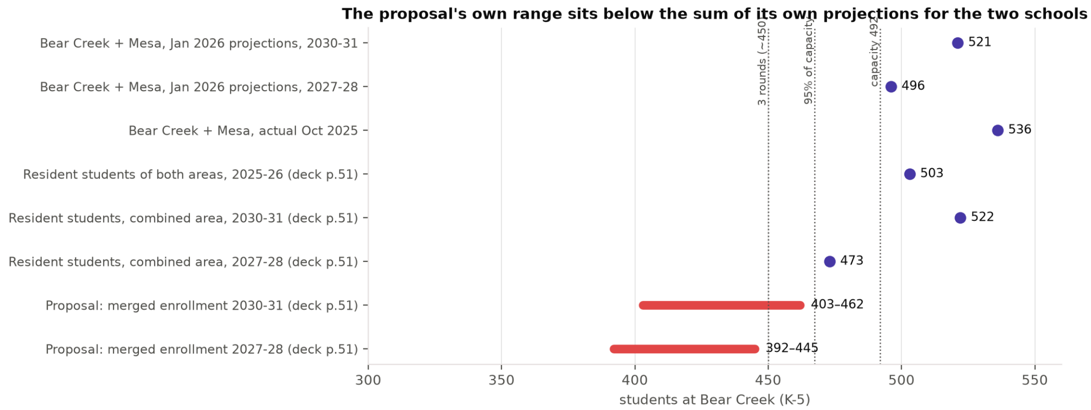
School	Target year	Jan 2024 run	Jan 2025 run	Jan 2026 run	Actual
Bear Creek	2024–25	287	318	–	318
	2025–26	273	308	312	312
	2026–27	271	302	299	
	2027–28	262	290	293	
	2028–29	248	274	288	
	2029–30	–	272	310	
	2030–31	–	–	320	
Mesa	2024–25	220	230	–	230
	2025–26	211	230	224	224
	2026–27	202	220	217	
	2027–28	203	218	203	
	2028–29	202	216	203	
	2029–30	–	224	202	
	2030–31	–	–	201	

1. The October 2025 work session, which introduced the Board to the South Boulder question, contained no new projections: its slide 17 is number-for-number the January 2025 table.

2. Between the January 2025 and January 2026 runs, Bear Creek's 2029–30 projection rose by 38 students (+14%) and Mesa's fell by 22 (−10%). Bear Creek's 2028–29 projection rose in each of three successive runs, from 248 to 274 to 288 (+16% over two years).

3. The January 2026 path for Bear Creek is the first to turn upward (299, 293, 288, then 310, 320). The documents reviewed do not say why. The one boundary change adopted for 2026–27 converts the Bear Creek/Creekside dual area to Bear Creek only; that area holds 37 elementary students, 23 of whom already attend Bear Creek (Sept. 9, 2025 boundary study p. 22), so it does not account for a 38-student revision.

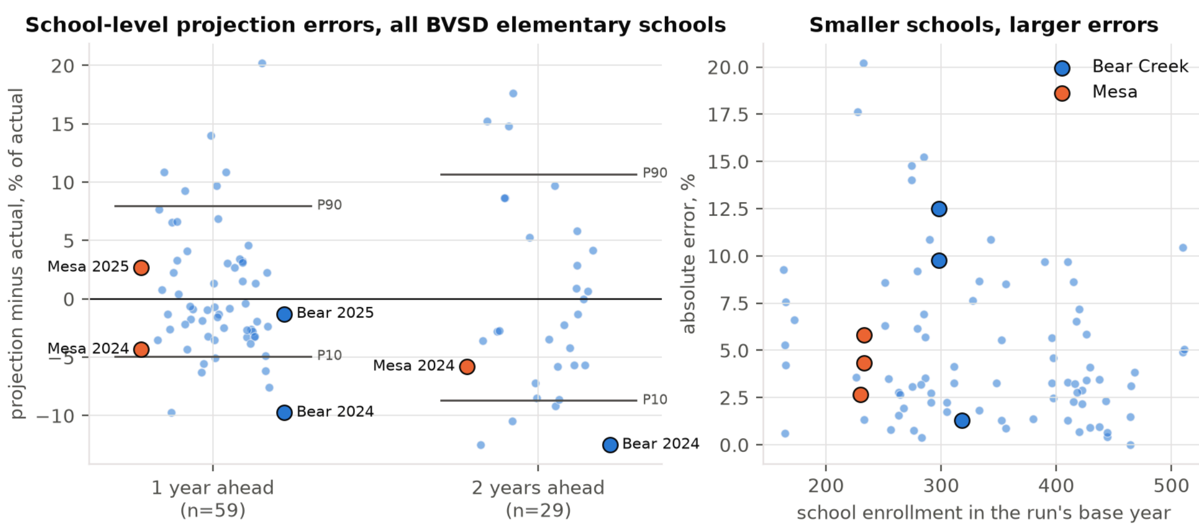
4. The proposal's merged-school range sits below the sum of the district's own separate projections (Figure 2): 521 in 2030–31 and 496 in 2027–28 against ranges topping out at 462 and 445.



The proposal's post-merger range against the district's own separate projections, resident counts and thresholds.

How accurate have the district's school-level projections been?

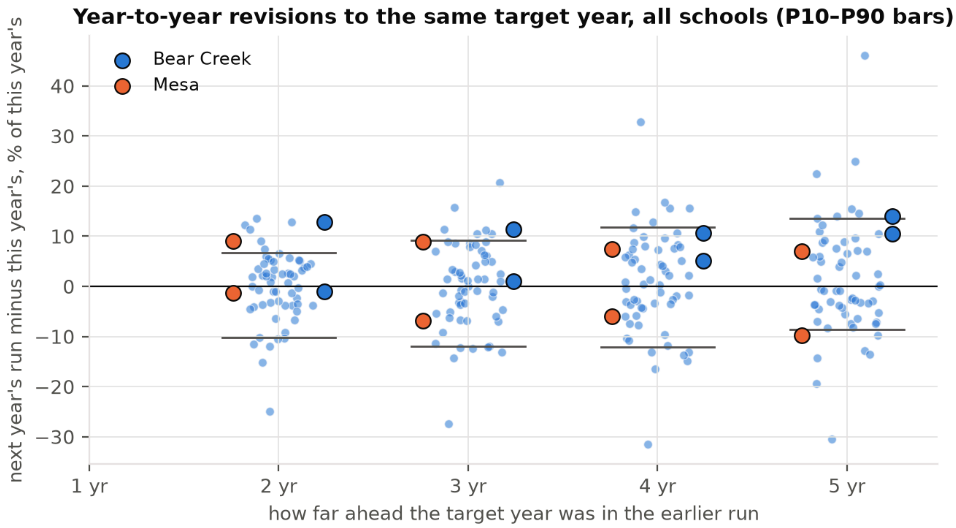
With three runs and official counts through October 2025, the projections can be scored one and two years ahead for every elementary school: 59 one-year cases and 29 two-year cases (mountain schools excluded). Figure 3 and Table 3 summarize the errors; Figure 4 shows revisions between successive runs to the same target year, the only available evidence on uncertainty three to five years out.



Left: percent error of BVSD projections by horizon, all elementary schools, Mesa and Bear Creek highlighted. Right: absolute error against school size.

Absolute percent error of BVSD school-level projections (all elementary schools except Gold Hill and Jamestown) and magnitude of revisions to the same target year between successive runs.

	n	Median	80th pct.	90th pct.	Max	Mean signed
Error, 1 year ahead	59	3.2%	6.4%	9.4%	20.2%	+0.6%
Error, 2 years ahead	29	5.7%	9.4%	13.0%	17.6%	0.0%
Error, 2 years ahead, schools under 300	13	5.8%	13.9%	–	–	–
Revision, target 2 years out	59	3.9%	9.1%	11.5%	24.9%	
Revision, target 3 years out	59	6.0%	10.8%	12.3%	27.4%	
Revision, target 4 years out	59	6.2%	12.2%	15.1%	32.8%	
Revision, target 5 years out	59	6.5%	12.5%	14.7%	46.1%	



Percent revision between successive runs to the same target year, by how far out the target was in the earlier run.

Three points bear on the decision. First, district-wide totals are projected well (errors of −1.1%, −0.7% and +1.4% for the three total-level cases; the district's own Dec. 2025 update reports the 2025–26 count came in "1% more than projected", news article p. 1). The school-level errors are therefore not a sign of a poor model; they are what school-level cohort projections look like when resident moves, open enrollment and small numbers all contribute. Second, errors are larger for small schools, and Mesa is among the smallest. Third, the mean signed error is near zero: the district has been about as likely to be too low as too high, which is the situation in which an irreversible choice is costly.

For the two schools at issue, the January 2024 run projected Bear Creek at 287 for October 2024 and 273 for October 2025; the actuals were 318 and 312 (−9.7% and −12.5%). Mesa was projected at 220 and 211 against actuals of 230 and 224. The January 2025 run did well one year out (Bear Creek −1.3%, Mesa +2.7%).

Small-sample caution. The two-year 90th percentile rests on 29 cases and should be read as "roughly one in ten misses by more than 13%", not as a precise quantile. The backtest coverage figures in Section 6 (75% and 73% against a nominal 80%) are based on 52 and 26 cases and are statistically indistinguishable from nominal.

An independent projection with stated uncertainty

Method

The model follows the district's own method (cohort survival, Feb. 2026 report p. 7) but reports a distribution. For each school, twelve years of October K–5 counts give eleven annual grade-progression ratios for grades 1–5 and twelve kindergarten intakes. Future years are simulated 10,000 times; in each simulated year one historical year is drawn at random and its kindergarten residual and all five progression ratios are applied jointly to both schools, so that a shock like 2020 stays one event and the schools stay correlated.

Kindergarten intake is the one place where a modelling choice is unavoidable, and it is made two ways, presented as co-equal specifications throughout:

Model A-trend.

Log-linear trend in kindergarten intake fitted to 2014–2025, with parameter uncertainty (pairs bootstrap). This extrapolates the twelve-year decline.

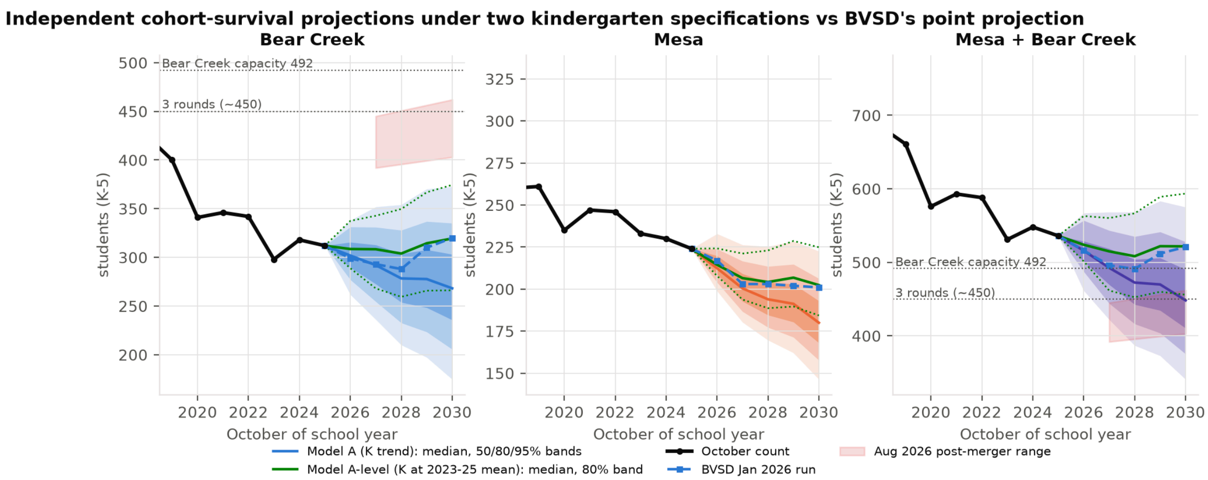
Model A-level.

Kindergarten intake held at its 2023–25 mean with the same year-to-year noise. This assumes the decline has ended.

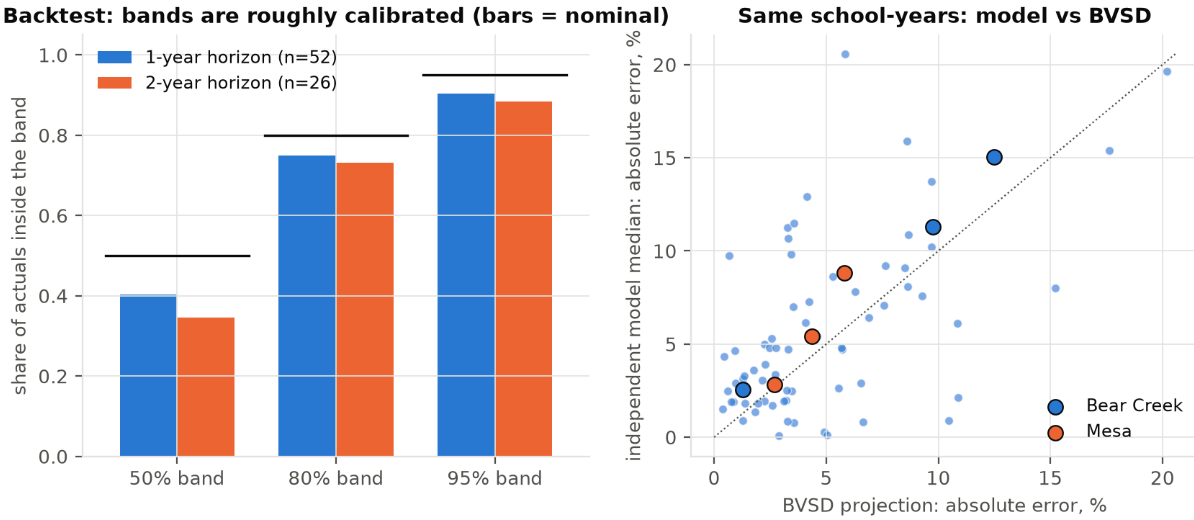
A random walk with bootstrapped drift on log totals (Model B) is run as a further check. Scripts: `analysis/03_independent_projection.py`.

Backtest

Fitting on data through October 2023 and October 2024 and scoring on the subsequent counts for all 26 elementary schools with complete records (Figure 6): the nominal 80% band contained 75% of one-year actuals (n = 52) and 73% of two-year actuals (n = 26); the 95% band, 90% and 88%. The bands are calibrated within sampling error. The model's median errors (3.4% at one year, 8.3% at two) are of the same order as the district's on the same school-years (3.3% and 5.7%); it is not a better forecaster than the district and is not offered as one. It is a calibrated way of asking how wide the band is.



Independent projections under the two kindergarten specifications (Model A-trend shaded 50/80/95%; Model A-level median and 80% band dotted), with BVSD's January 2026 run and the proposal's post-merger range.



Backtest across all schools: coverage of the model's bands, and its errors against BVSD's on the same school-years.

Results

The independent model's median at 90% retention (430, Model A-trend) lies inside the district's 403–462 range. That is not the finding; the district's central estimate is not in dispute. The finding is in Table 4 and Section 7: the probability mass above 450 and above 492, and the sensitivity of everything to the kindergarten specification.

Independent projection for 2030–31 (students, K–5) under the two kindergarten specifications and the check model.

	10th pct.	25th	Median	75th	90th
Bear Creek, A-trend	206	236	268	303	335
Bear Creek, A-level	266	291	319	349	374
Mesa, A-trend	158	168	180	193	206
Mesa, A-level	184	192	202	214	225
Both schools, A-trend	375	410	448	490	528
Both schools, A-level	456	487	522	559	594
Both schools, Model B	404	431	463	498	533
BVSD Jan 2026 run, both schools			521		
Proposal range, merged school		403		462	

Two kindergarten specifications, and why the choice decides the answer

Combined kindergarten intake at the two schools was 105, 105, 95, 102, 86, 79 in 2014–2019, stepped down to 62 in 2020, and has been 82, 77, 58, 84 and 76 since (October pupil-count files, grade columns; Figure 10). A trend fitted through a step-then-plateau extrapolates a decline that, on this evidence, ended five years ago; a level fitted to the plateau assumes it will hold. Two district statements favour the level reading: the Dec. 12, 2025 update reports that "the number of births in Boulder seems to be leveling after a steep decline until 2019" and that "kindergarten enrollment came in higher than projected, and for the second year in a row, the rate of new kindergarteners showing up was 93% of the number of births five years prior" (news article p. 1); the Feb. 2026 report assumes the births-to-kindergarten ratio is "relatively stable" (p. 8). Neither settles the question, and the district's own Bear Creek path (rising to 320) is closer to the level specification than to the trend one.

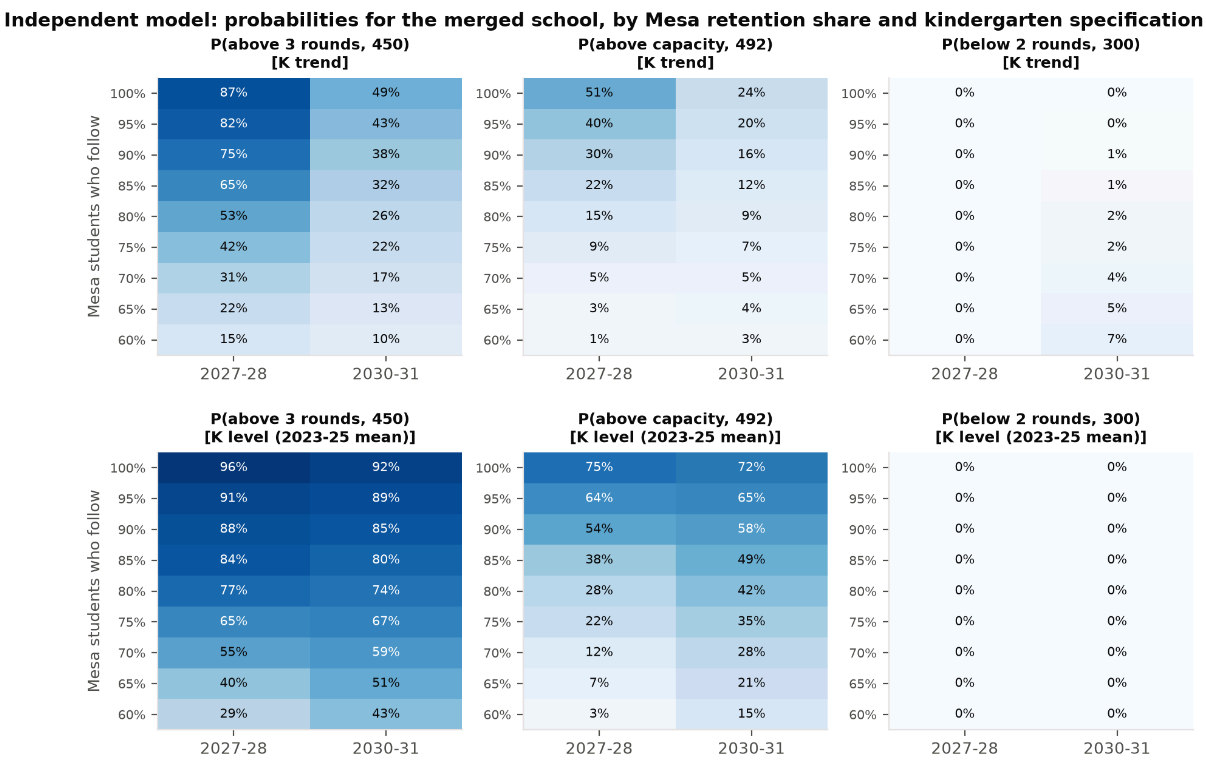
What matters for the Board is the size of the gap. At 90% retention, the merged school's 2030–31 median is 430 under A-trend and 501 under A-level: 71 students, or about three classrooms, more than the 59-student width of the district's entire published range. The district's documents do not say which assumption its own projection makes. Whether the merged school fits the building therefore turns on an undisclosed specification choice.

Merged school in 2030–31 by kindergarten specification, 90% of Mesa students following (independent model).

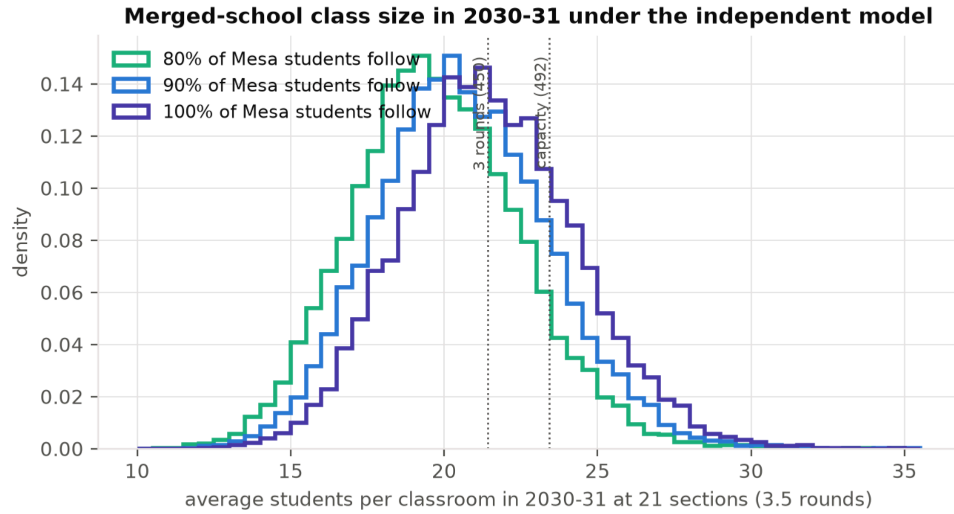
Specification	Median	10th pct.	90th pct.	P(> 450)	P(> 492)	P(< 300)
A-trend	430	359	508	38%	16%	1%
A-level	501	437	572	85%	58%	0%

Consolidation scenarios

Let r be the share of Mesa's students who enrol at Bear Creek rather than elsewhere. Merged enrollment is Bear Creek's simulated path plus r times Mesa's, drawn jointly. Figure 7 gives the probabilities that matter under both specifications; Figure 8 gives implied class sizes at 21 sections (3.5 rounds across six grades).



Probability that the merged school exceeds three rounds, exceeds 492, or falls below two rounds, by share of Mesa students who follow; top row A-trend, bottom row A-level.



Distribution of average class size in 2030–31 at 21 sections (A-trend).

Under A-trend with 90% following, the 2030–31 median is 430 (80% band 359–508), the probability of exceeding 450 is 38%, of exceeding 492 is 16%, and of falling below 300 is 1%. Under A-level the median is 501 (437–572), and the probabilities are 85%, 58% and 0%. In the first year of the merged school, 2027–28, the medians are 473 and 494 and the probabilities of exceeding 492 are 30% and 54%. The P(< 300) column shows that the downside of a too-small merged school is negligible under either specification; the risk is asymmetric toward overcrowding.

The proposal's own range can be decoded the same way. Against the district's separate January 2026 projections (320 and 201 for 2030–31; 293 and 203 for 2027–28), the ranges 403–462 and 392–445 imply that r lies between 0.41 and 0.71 in 2030–31 and between 0.49 and 0.75 in 2027–28: the district's upper bound assumes that between a quarter and three-fifths of Mesa's projected students go elsewhere. If they do not, the district's own numbers put the merged school above the building's capacity.

RISE, AIM and effective capacity

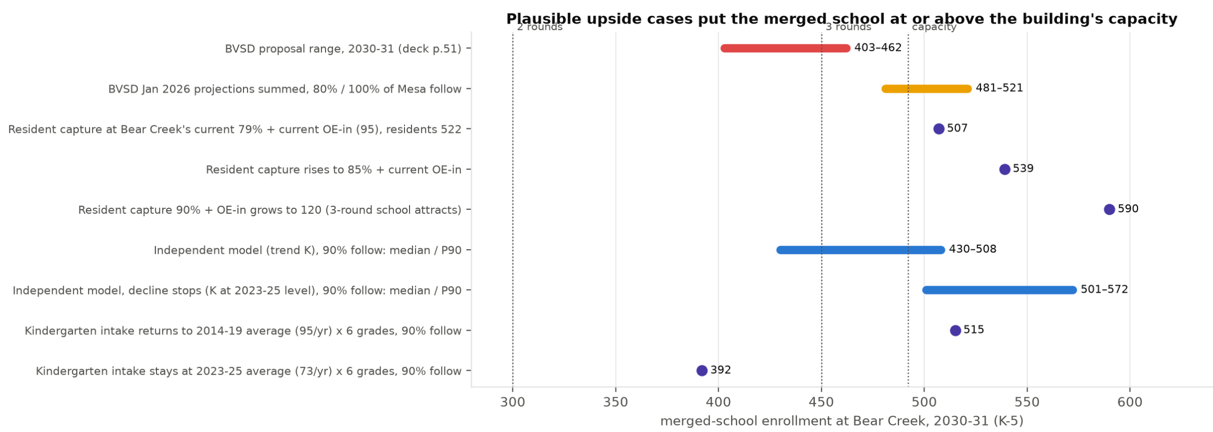
Bear Creek's 492 seats are "program capacity based on current use of the building" (Feb. 2026 report p. 9, footnote). The bvsd.org proposal page states that "all existing AIM and RISE programs would be housed at Bear Creek" and the deck that "Mesa's RISE Program moves to the Bear Creek building" (page p. 7 of the saved copy; deck p. 49). Center-based programs for students with autism and affective needs run at low ratios and occupy classrooms. None of the documents reviewed states how many rooms these programs occupy at either school, or whether 492 nets them out. If it does not, general-education capacity is lower than 492. Table 6 recomputes the exceedance probabilities at effective capacities of 492, 470 and 445 (roughly one and two classrooms removed); it is illustrative pending the district's statement of what 492 assumes.

Probability that merged enrollment exceeds an effective general-education capacity (independent model; 90% and 100% of Mesa students following).

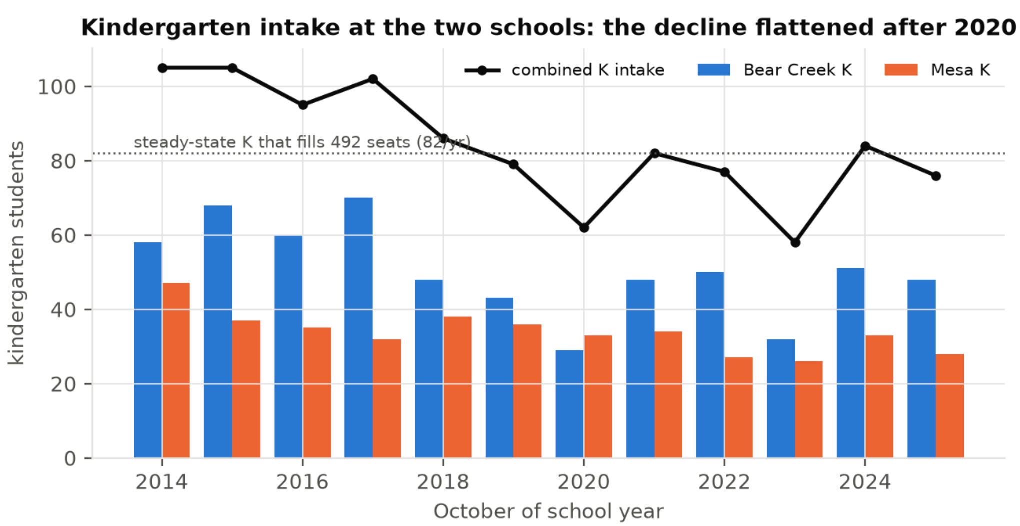
Specification	Year	Follow	Median	P(> 492)	P(> 470)	P(> 445)
A-trend	2027–28	90%	473	30%	53%	79%
A-trend	2027–28	100%	492	51%	75%	90%
A-trend	2030–31	90%	430	16%	26%	41%
A-trend	2030–31	100%	448	24%	37%	52%
A-level	2027–28	90%	494	54%	77%	88%
A-level	2027–28	100%	515	75%	88%	97%
A-level	2030–31	90%	501	58%	74%	87%
A-level	2030–31	100%	522	72%	85%	93%

What would it take for Bear Creek to fill? Upside cases

The district attributes the decline to low births, an aging population and constrained housing turnover (FAQ p. 8; deck p. 10). Each driver can also reverse, and the merged school starts close to the thresholds. Figure 9 lays out named cases for 2030–31; Figure 10 shows the kindergarten series.



Merged-school enrollment in 2030–31 under the proposal, under the district's own projections, and under named upside cases.



Kindergarten intake at the two schools, 2014–2025 (October pupil-count files).

Leveling-off.

> A building of 492 seats is filled in steady state by an intake of 82 a year; the 2023–25 mean is 73, the 2014–19 mean 95, and 2024's intake was 84. Model A-level is this case with noise: median about 500, 90th percentile about 570.

Resident capture.

> Today 79% of Bear Creek's resident students attend Bear Creek and 68% of Mesa's attend Mesa (deck pp. 44 and 51). The district projects 522 residents in the combined area in 2030–31 (p. 51). At Bear Creek's current capture rate, plus Bear Creek's current 95 open-enrolled-in and out-of-district students, the merged school would have about 507; at 85% capture, about 539. The proposal's 403–462 is consistent with those inputs only if capture falls to 59–70%, below either school's rate today.

Open enrollment.

> The proposal's stated aim is that a three-round school with fuller programs is a better student experience (deck pp. 60–61). If so, it will attract open enrollees; every 25 is one classroom. Bear Creek currently has 82 in-district open-enrolled students and 13 out-of-district students (derived from deck p. 44 and the 2025–26 special-programs file); the district does not publish which schools they come from.

Housing turnover.

> The district's boundary study uses a yield of 58 elementary students per 324 single-family dwellings, 0.18 per home (Sept. 9, 2025, p. 22). Moving the A-trend median of 430 to 492 requires about 62 students: about 350 homes turning over at that average yield, or 62 homes passing from households without children to households with one elementary-age child. The FAQ describes the over-60 population as the fastest-growing segment and "aging in place" as a constraint on turnover (p. 8); that is a statement about timing, and the same households will eventually sell. No age-structure or housing data for the two areas is in the repository, so this case is parametric.

The district's own numbers, unadjusted.

> If the January 2026 projections hold and all of Mesa's students follow, the merged school has 521 students in 2030–31, 29 above capacity.

Portfolio interdependence

Every projection in this appendix, the district's and ours, is fitted on grade-progression ratios and open-enrollment flows observed under the current Boulder portfolio and the current choice rules. The proposal changes both in 2027–28. It closes Douglass, Flatirons and Birch and the Community Montessori building; relocates High Peaks to the Douglass campus and Community Montessori to the Aurora 7 building with BCSIS; redraws attendance areas across the region (deck pp. 46, 52, 23, 59); and introduces a one-year open-enrollment priority for students of closed or moved schools (deck p. 63; FAQ p. 2, "The Board will study this item further on Sept. 8"). Fitted ratios do not carry across a change of that size.

Where the closed schools' students are assigned (from the deck). Douglass's area is divided among Coal Creek, Eisenhower and Heatherwood (p. 46), with post-change ranges of 395–427, 297–302 and 260–268 in 2030–31 (p. 48). Flatirons' area is split at Boulder Canyon Drive between Foothill (north) and Whittier (south) (p. 52), with post-change ranges of 484–492 and 315–329 (p. 54). Birch's area goes to Kohl, 437–474 (p. 25). No part of the Flatirons area is assigned to Bear Creek; the deck assigns none of the closed schools' residents to Bear Creek other than Mesa's. After the package, the Boulder schools the deck projects at or near three rounds in 2030–31 are Foothill (484–492), Bear Creek (403–462 at its upper end) and Coal Creek (395–427); district-wide, the deck reports the number of schools below two classes per grade falling from 14 to 6 and average classes per grade rising from 2.0 to 2.5 (p. 58). Among the options studied and not proposed was closing Creekside and redistributing its students "to Mesa, Bear Creek and Eisenhower" (p. 56).

What the documents say about re-sorted flows. The FAQ states that where "new short term enrollment will push the school near capacity, enrollment projections take into account managing choice enrollment beginning in 2027–28 to ensure that resident students can be accommodated" (p. 3). Neither the deck nor the FAQ states whether the 403–462 range accounts for open-enrollment flows re-sorted from other schools in the package, and no document gives Bear Creek's sending schools.

Direction. The identifiable channels point toward higher enrollment at Bear Creek: it becomes one of three Boulder schools at or near three rounds; the proposal's own claim is that fuller programs attract enrollment; and choice priority is being extended to displaced families across the region. The direction is not certain and the district has not modelled it. No flow assumptions are invented here and no row is added to Figure 9 for this channel, because the deck gives no count of residents reassigned to Bear Creek beyond Mesa's.

Tie to the standard. A projection cannot be back-tested against a portfolio that has not yet existed. Standard (i) therefore cannot be met for the merged school until at least one October count under the new portfolio exists.

Other sources of variance not in any projection

Each item is listed with its documentary support; where none exists it is labelled unquantified.

- **Announcement effect.** Open enrollment for 2027–28 runs Nov. 1 to Dec. 18, 2026, after the Sept. 22 vote (deck p. 67). Mesa families who leave in that window lower the 2026–27 Mesa base to which the district's retention range is applied. Magnitude unquantified in any document.

- **Compressed choice window.** The window "has been adjusted to end prior to Winter Break" (deck p. 67). Effect on flows unquantified.

- **A charter or other operator in a vacated building.** "Decisions about the future use of buildings are not part of the Resilient Schools proposal" (FAQ p. 3). Press reporting on district emails (Boulder Reporting Lab, Aug. 20, 2026; not in the repository, see `data/SOURCES.md` L2) has raised the possibility; unquantified.

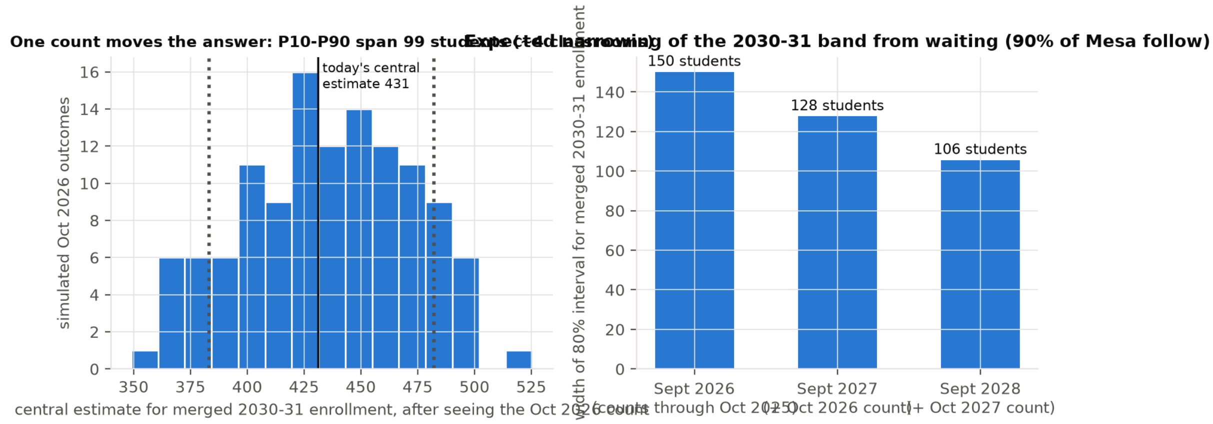
- **Housing turnover in an aging stock.** FAQ p. 8 (see Section 8). Unquantified.

- **Kindergarten capture under universal preschool.** CDE records 13, 20 and 25 pre-kindergarten students at Mesa in 2023–24 to 2025–26 that BVSD's tables omit; the proposal says preschool "will continue to be offered in the area" without a location (deck p. 64). No document addresses the effect of preschool placement on kindergarten capture. Unquantified.

- **The 2026–27 boundary change.** 37 students in the dual area, 23 already at Bear Creek (Sept. 9, 2025 study p. 22). Small and documented, but not yet observed in any count.

What the next October count would change

Re-running the model after each of 120 simulated October 2026 outcomes shows the central estimate for merged 2030–31 enrollment moving across a range of about 100 students from the 10th to the 90th percentile of those outcomes (Figure 11, left panel): a single count could move the answer by roughly two classrooms in either direction. The expected width of the 80% band also narrows, from about 150 students to about 128 (right panel), but the swing in the centre is the larger effect. The October 2026 count is the first that will be taken under the 2026–27 boundaries, and the January 2027 run the first that could reflect them. Standard (i) cannot be met without it.



Left: central estimate for merged 2030–31 enrollment after each of 120 simulated October 2026 counts. Right: expected width of the 80% band by decision date.

Cost and benefit

The district's figures (deck p. 60, for the whole package): recurring annual savings of \$3.5–4.0M, "estimated funding redirected to support the student experience every year"; one-time transition costs of \$7.5–10.0M for facility modifications ("some offset by savings from Bond projects") and \$5.0M for school transition support ("budgeted, FY27"). No per-school breakdown, no figure for maintaining vacant buildings, no portable cost, no RISE or AIM relocation cost and no transportation cost appears in the deck, the FAQ or the bvsd.org pages reviewed. Two one-time figures circulate publicly: "up to \$10M" is the facility line alone and "\$12.5–15M" is the sum of both lines; both are consistent with p. 60 (`data/CONFLICTS.md` C11).

Arithmetic on the district's figures. Simple payback of the one-time cost from the recurring savings is 3.1 to 4.3 years (12.5–15.0 divided by 3.5–4.0). If the general fund is about \$494M, a figure that circulates publicly but appears in none of the documents reviewed, the recurring savings are 0.7–0.8% of it. There is no Mesa-specific line on either side.

Realization risk. The benefit is a projection with realization risk; the one-time cost is undercounted if the merged school needs portables or further boundary work. On the revenue side, a statewide study of California districts (Pearman, Stanford SCALE, May 2026; about 800 districts, 2011–2019) reports that closures cut spending by about \$447 per student but cut revenue by about \$433 per student as families left, with no significant improvement in the probability of a balanced budget and no significant reduction in teachers, principals or total staff. (Figures as reported; the paper is not in the repository and must be verified before citation, `data/SOURCES.md` L1.) The mechanism is the one the district's own range assumes: 403–462 implies that 29–59% of Mesa's projected students do not follow to Bear Creek. Mesa is 0.71 miles from Bear Creek (deck p. 49), so leakage out of the district is likely below the California average, but the district's own range assumes some, and the deck does not say where those students go or what revenue follows them.

Other gaps: questions the documents do not answer

1. How many classrooms do the AIM and RISE programs occupy at Bear Creek today and after Mesa's RISE program moves, and does 492 net them out? (deck p. 49; bvsd.org page p. 7)

2. School-age care "will be offered at Bear Creek" (bvsd.org page p. 7); with what capacity relative to two schools' current demand?

3. Preschool "will continue to be offered in the area"; where? (deck p. 64; FAQ p. 4: locations "will be shared by Oct. 1", after the vote.)

4. Transportation: elementary eligibility is 1.5 miles or more from the assigned school (deck p. 64); how many Mesa-area addresses fall inside and outside that radius for Bear Creek?

5. The K–2 / 3–5 reconfiguration of Mesa and Bear Creek is listed among "other options studied" (deck p. 56); in October its fiscal impact was described as "uncertain" (work session slide 45). No scoring of it against the adopted option has been published.

Limitations

Three projection runs exist, so direct accuracy is measured only one and two years ahead; the three-to-five-year figures are revisions, a lower bound on uncertainty. The independent model uses only BVSD's own grade counts; it has no births, housing or open-enrollment inputs. Retention r and the open-enrollment response are assumptions shown as ranges. The effective-capacity table is illustrative. The turnover case uses a district yield figure from a different neighbourhood and no local housing data. Costs of either error are not quantified because the district's documents do not quantify them. Capacity is the district's "program capacity based on current use of the building" and is itself a policy variable. The Pearman figures are cited as reported and not yet verified against the paper.

Conclusion

The documents presented do not contain a projection with prediction intervals, do not state the retention, capture or open-enrollment assumptions behind the merged-school range, do not show that the merged school fits the building under upside cases net of program space, contain no Mesa-specific cost or revenue figures, and leave the transition questions in Section 13 open. The Board should decline to approve the Mesa/Bear Creek component of the Resilient Schools proposal at this time, and direct staff to return with an analysis that meets this standard.

Reproducibility

All scripts run from the repository root in numbered order: `scripts/` (extraction and verification) and `analysis/01_descriptive.py` through `06_value_of_waiting.py`. Figures are written to `figures/`, tables to `analysis/output/`. Random seeds are fixed. The source of this document is `report/report.tex`; `report/build.sh` renders it.

Verification summary

Every figure the argument relies on was checked against a primary page: capacities (Oct. 2025 slide 17; Feb. 2026 p. 9); the 2029–30 projections in both runs; the 403–462 range (deck p. 51); the thresholds (slide 8; Feb. 2026 p. 10); the cost figures (deck p. 60); the boundary-area counts (Sept. 9, 2025 study p. 22); and all twelve years of counts for both schools (pupil-count files, grade sums and prior-year columns reconciling exactly). The full log is `data/VERIFICATION.md`.

Scenario tables

Merged-school outcomes in 2030–31 by share of Mesa students who follow, both specifications.

Spec.	Share	Median	10th	90th	P(> 450)	P(> 492)	P(< 300)
A-trend	80%	412	342	489	26%	9%	2%
A-trend	90%	430	359	508	38%	16%	1%
A-trend	100%	448	375	528	49%	24%	0%
A-level	80%	481	418	549	74%	42%	0%
A-level	90%	501	437	572	85%	58%	0%
A-level	100%	522	456	594	92%	72%	0%